

-- Cox Regression - Sarcopenia Onset -----

i Definition : ewgsop2  
i Covariates : time\_dependent  
i Dairy : categorical  
i Route : mice  
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.  
Fitting .imp = 1 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years  
Fitting .imp = 2 / 20 ...  
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years  
Fitting .imp = 3 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1949 intervals | 52 events | 7804.1 person-years  
Fitting .imp = 6 / 20 ...  
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years  
Fitting .imp = 7 / 20 ...  
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years  
Fitting .imp = 8 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years  
Fitting .imp = 9 / 20 ...  
Events: 56 sarcopenia cases out of 1121 participants

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 52 events | 7801.6 person-years  
Fitting .imp = 11 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years  
Fitting .imp = 12 / 20 ...  
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years  
Fitting .imp = 13 / 20 ...  
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years  
Fitting .imp = 14 / 20 ...  
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years  
Fitting .imp = 17 / 20 ...  
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years  
Fitting .imp = 18 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years  
Fitting .imp = 19 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy\_exposure levels: "Q1", "Q2", "Q3", and "Q4"  
??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years  
Fitting .imp = 20 / 20 ...  
Events: 56 sarcopenia cases out of 1121 participants

```
-- Events Per Variable (EPV) [mice (all imputations)]
```

```
i Events : 49
```

```
i Coefs : 10
```

```
i EPV : 4.9
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-- C-index (Harrell's concordance) [mice (averaged)]
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```
i C-index : 0.786 (95% CI: 0.731-0.842)
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```
i SE : 0.0285
```

```
??? C-index >= 0.70 - good discrimination.
```

```
-- Global model tests [mice (averaged)]
```

```
i Likelihood ratio test :  $\chi^2(10) = 59.13$ ,  $p = 7.35e-09$  <- preferred
```

```
i Wald test :  $\chi^2(10) = 43.89$ ,  $p = 4.05e-06$ 
```

```
i Score (log-rank) test :  $\chi^2(10) = 44.98$ ,  $p = 2.62e-06$ 
```

```
i Total person-years (mean across imputations): 7801.7
```

```
-- Assumption checks [mice_imp1] --
```

```
-- 1. Proportional Hazards - cox.zph() | pass: global  $p > 0.05$ 
```

```
chisq df p dairy_exposure 3.4458 3 0.33 BMI 0.0875 1 0.77 education_level
```

```
3.1688 2 0.21 smoking_status 3.9312 2 0.14 mvpa_min_day_f1 0.2729 1 0.60
```

```
diabetes_status 2.1074 1 0.15 GLOBAL 14.3536 10 0.16
```

```
??? PH assumption not violated (global  $p = 0.1575$ )
```

```
Schoenfeld plots saved to
```

```
03_outputs/Cox/mice_ewgsop2_time_dependent_categorical_27072026_1254
```

```
-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear
```

```
'geom_smooth()' using formula = 'y ~ x'
```

```
'geom_smooth()' using formula = 'y ~ x'
```

```
Linearity plots saved to
```

-- 7. Log-log survival plot | pass: parallel lines across log(time)

Log-log plot saved to

03\_outputs/Cox/mice\_ewgsop2\_time\_dependent\_categorical\_27072026\_1254

-- 8. Cox-predicted vs KM - calibration by dairy quartile

Cox vs KM calibration plot saved to

03\_outputs/Cox/mice\_ewgsop2\_time\_dependent\_categorical\_27072026\_1254

-- Kaplan-Meier plot by dairy category [mice\_imp1]

i KM (td): baseline dairy quartile x final event status (49 events in 1121  
pts).

Ignoring unknown labels:

o colour : "Dairy quartile"

i Log-rank p-value: 0.596

KM plot saved to

03\_outputs/Cox/mice\_ewgsop2\_time\_dependent\_categorical\_27072026\_1254

-- Adjusted scenario survival curves (time-dependent) [mice\_imp1]

-- Incidence rates by dairy quartile [mice]

quartile n\_pts n\_events person\_years rate\_per\_1000py ci\_low ci\_high 1 Q1 271

10.00 1894.20 5.28 2.53 9.71 2 Q2 291 10.70 2044.03 5.23 2.58 9.44 3 Q3 272

12.00 1879.40 6.39 3.30 11.15 4 Q4 287 17.45 1984.10 8.79 5.16 14.00

Incidence table saved to

03\_outputs/Cox/mice\_ewgsop2\_time\_dependent\_categorical\_27072026\_1254

Model results saved to

03\_outputs/Cox/mice\_ewgsop2\_time\_dependent\_categorical\_27072026\_1254/mice\_model\_results.csv